

1)  $p(A) = \frac{m}{n}$  — число благоприятных исходов / — число всех возможных исходов  $m=12$   $n=8$   $p(A) = \frac{8}{12} = \frac{2}{3}$  Ответ:  $2/3$

2) 

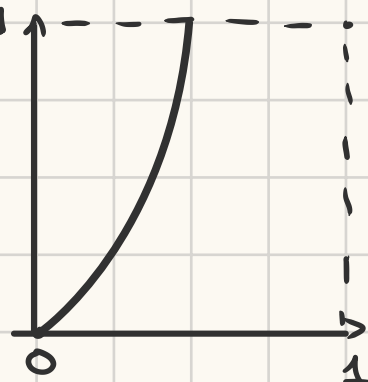
|       |       |
|-------|-------|
| 5 Сер | 2 Чер |
|-------|-------|

 $n = C_7^2$   $p(A) = \frac{10}{21}$   $m = C_5^2$  Ответ:  $10/21$

3)  $n = C_8^3 = 56$   $m = C_5^2 \cdot C_3^1 = 30$   $p(A) = \frac{30}{56} = \frac{15}{28}$  Ответ:  $15/28$

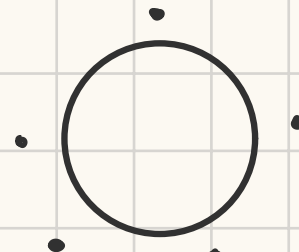
7) 20x | 3 Грус | 5 Шк  $n = C_2^1 \cdot C_3^2 \cdot C_5^2$   $m = C_{10}^5$  Ответ:  $p(A) = \frac{C_2^1 \cdot C_3^2 \cdot C_5^2}{C_{10}^5}$

10)  $x^2 + px + q = 0$   $D = p^2 - 4q$   $p(A) = \frac{\mu(A)}{\mu(\Omega)}$   $\mu(\Omega) = 1$   
 а)  $D > 0$   $p^2 - 4q > 0$   $1$   
 $p^2/4 > q$   $\mu(A) = \int_0^1 \frac{p^2}{4} dp = \frac{1}{12}$   $p(A) = \frac{1/12}{1} = \frac{1}{12}$



б)  $D = 0$   $p^2 = 4q$  — касание  $\mu(A) = 0$   $p(A) = \frac{0}{1} = 0$  Ответ: а)  $\frac{1}{12}$  б) 0

12) —————  $n = 8!$   $m = \overbrace{6 \cdot 6!}^{\text{ограничение}} = 6 \cdot 6!$  Ответ:  $p(A) = \frac{3}{28}$

13)   $n = 4!$   $m = 3! \cdot 2!$   $p(A) = 1/2$  Ответ:  $p(A) = 1/2$

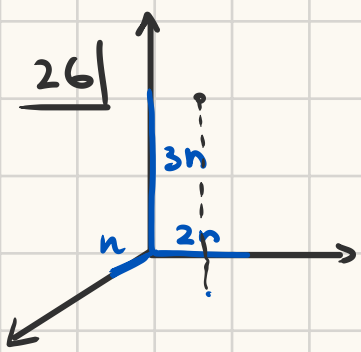
14)  $n = 5^3$   $p(A) = \frac{12}{25}$  Ответ:  $p(A) = 12/25$   $m = A_5^3$

15)  $n = 365^{25}$   $m = 365^{25} - A_{365}^{25}$    $A_{365}^{25}$   $365^{25}$   $p(A) = 1 - \frac{A_{365}^{25}}{365^{25}} \approx 0.57$  Ответ: 0,57

18)  $n = C_{52}^6$   $p(A) = \frac{m}{n} \approx 0.43$  б)  $m = 4 \cdot C_{13}^5 \cdot C_{39}^1$   $p(A) = \frac{4 \cdot C_{13}^5 \cdot C_{39}^1}{C_{52}^6} \approx 0.00986$   
 а) 

|   |   |   |   |
|---|---|---|---|
| 3 | 1 | 1 | 1 |
| 2 | 2 | 1 | 1 |

 $\left. \begin{array}{l} \text{варианты} \\ \text{но} \\ \text{мастям} \end{array} \right\}$  Ответ: а) 0,43  
 $m = 4 \cdot C_{13}^3 \cdot (C_{13}^1)^3 + C_4^2 \cdot (C_{13}^2)^2 \cdot (C_{13}^1)^2$  б) 0,01

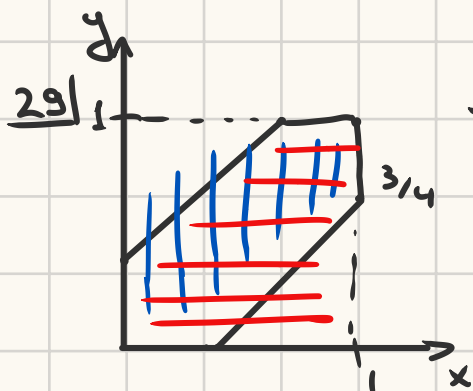


$$m=6 \quad n = \frac{(6n)!}{(n!)(2n!)(3n!)} = C_{6n}^n \cdot C_{5n}^{2n} \cdot 1$$

$$p(A) = \frac{m}{n} = \frac{(n!)(2n!)(3n!) \cdot 6}{6n!} \quad \text{Ombet: } \frac{(n!)(2n!)(3n!) \cdot 6}{6n!}$$

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$$n = C_{45}^5 \quad m = C_{k-1}^4 \quad \text{Ombet: } p(A) = \frac{C_{k-1}^4}{C_{45}^5}$$

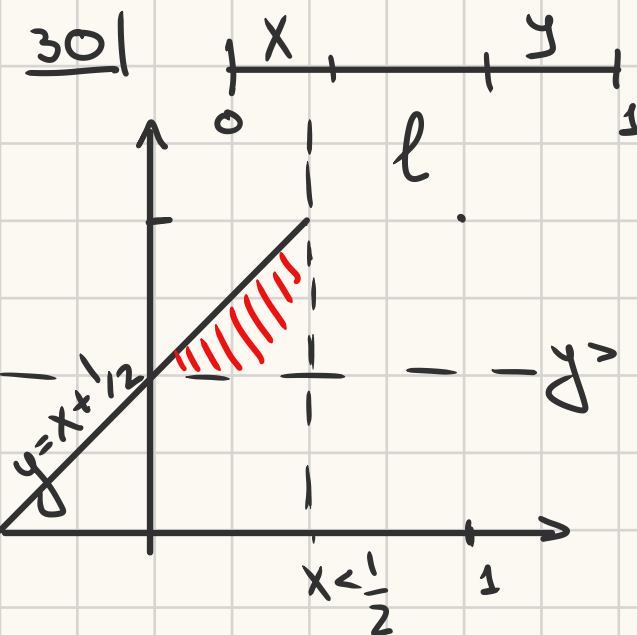


$$\Omega = [0,1] \times [0,1] \quad \text{15 мин - 1/4 часа} \quad |x-y| \leq 0.25 \rightarrow \begin{cases} x - \frac{1}{4} \leq y \\ y \leq x + \frac{1}{4} \end{cases}$$

$$0 \leq x \leq 1 \quad 0 \leq y \leq 1$$

$$p(A) = 1 - 2 \cdot \frac{1}{2} (0.75)^2 = 1 - 0.5625 = 0.4375$$

$$\text{Ombet: } 0.4375$$



$$1) x \leq y; \quad x; y-x; 1-y - \text{группы частей}$$

$$\begin{cases} x < y-x+1-y \\ y-x < x+1-y \\ 1-y < x+y-x \end{cases}$$

$$\begin{cases} x < 1/2 \\ y-x < 1/2 \\ y > 1/2 \end{cases} \quad \mu(A) = 1$$

$$2) x \geq y$$

$$\mu(A) = \frac{1}{8}$$

случай 2

$$\text{Ombet: } p(A) = 1/4$$